Applied Ergonomics 94 (2021) 103398

Contents lists available at ScienceDirect

Applied Ergonomics

journal homepage: http://www.elsevier.com/locate/apergo

Navigating with Augmented Reality – How does it affect drivers' mental load?

Kassandra Bauerfeind [a,*], Julia Drüke [a], Jens Schneider [b], Adrian Haar [a], Lennart Bendewald [c], Martin Baumann [d]

[a] Volkswagen AG, Group Innovation, HMI Augmentation, Wolfsburg, Germany
[b] Volkswagen AG, Technical Development, Infotainment Services, Wolfsburg, Germany
[c] Volkswagen AG, Technical Development, HMI Domain Concepts, Wolfsburg, Germany
[d] University of Ulm, Institute of Psychology and Education Dept., Human Factors, Ulm, Germany

ARTICLE INFO

Keywords:
Augmented Reality
Head-up display
Driver's mental load
Ambiguous navigation task

ABSTRACT

Drivers have been proven to easily understand Augmented Reality (AR) information. Especially in an ambiguous navigation task, drivers are expected to benefit from AR information. The driving simulator study was aimed at examining differences in mental load while navigating in an urban area with ambiguous intersection situations (N = 59). The navigation information was presented to the driver through a head-up display (HUD): a conventional HUD or an AR display, which relates information to the surroundings. Additionally, the driver had to solve a non-driving-related task (NDRT) which was an auditory cognitive, spatial task. Results showed that while driving with the AR display, participants performed better in the NDRT, which indicates a reduced mental load compared with the HUD. Participants drove on average 3 km/h slower with the HUD, showing compensation behaviour.

1. Introduction

Navigating is an elementary driving task. It is a spatial task as the driver needs to orient himself in the environment. Drivers have to plan their routes and estimate costs and risks. For navigating in a foreign environment, or in ambiguous navigation scenarios, knowledge based behaviour (Rasmussen, 1983) is necessary. This requires the driver's resources and therefore can be cognitively demanding. However, it is possible to support the driver through improved displays. Because information and digitization in general, available traffic-related information in the vehicle is increasing, but information should be easy, readily accessible, and presented comprehensively without distracting the driver. For example, heads-up displays (HUD) enable drivers to receive driving related information within their field-of-view. Hence, information is readily available enabling concurrent scanning of the information provided and the environment (Prinzel and Risser, 2004) and minimizing drivers' gazes away from the street. However, the information presented in a HUD does not necessarily relate to the environment. Thus, drivers have to map the 2D information onto the real driving situation. This demands drivers' attention and may lead to navigation and driving errors.

Advanced HUDs are able to present Augmented Reality (AR) information, which is directly related to the driving situation. For example, while following the navigation arrow presented through an AR display, the driver gets the impression that the path-to-follow is virtually marked on the street. This might result in the driver easily and quickly understanding AR information. As the augmented information is correctly superimposed on the relevant objects in the environment, virtual and real objects are closer. Thus, the driver no longer has to switch between the relevant information and the real traffic situation (Kim and Dey, 2009). Moreover, with AR the driver does not have to map the virtual information onto the environment (Pfannmüller, 2017). Based on these advantages, AR is often applied in navigation. AR navigation information is intuitive to understand and facilitates the navigation task resulting in reduced navigation errors (Bauerfeind et al., 2019; Israel, 2012; Kim and Dey, 2009). In the Kim and Dey (2009) study, participants showed enhanced driving performance (e.g., completion time, number of missed turns) while driving with AR compared with digital maps shown on a center information display. This is in line with Medenica et al. (2011), who also found that participants showed an

* Corresponding author.
E-mail address: kassandra.bauerfeind@volkswagen.de (K. Bauerfeind).

https://doi.org/10.1016/j.apergo.2021.103398
Received 13 October 2020; Received in revised form 17 February 2021; Accepted 23 February 2021
Available online 12 March 2021

improved driving performance (i.e., lane position, steering wheel angle) while driving with an AR display for a navigation task rather than with a standard map-based navigation device or an egocentric street view navigation device. The egocentric street view navigation device was an LCD display presenting the driving situation from the driver's perspective including AR navigation information.

In the literature, several authors claim that AR navigation information requires less mental effort in understanding and interpreting information compared with navigation information in the HUD (Bengler et al., 2015; Israel, 2012; Kim and Dey, 2009; Pfannmüller, 2017; Pfannmüller et al., 2015). In the Medenica et al. (2011) study, participants felt less subjective workload while driving with navigation information presented in an AR display than any other navigation device, like map-based navigation. Most findings in the field of mental load levels with AR are based on subjective ratings. However, subjective data (questionnaire data) might be susceptible to distortions caused by social desirability. Furthermore, little is known about the effects on objective performance measures resulting from lower mental load level with AR. If the navigation task with AR (primary task) is less demanding, then the driver is more capable of performing better in other tasks (e.g., in a non-driving-related task [NDRT]) (Jahn et al., 2005). It must be closely observed whether AR information leads to reduced mental load resulting in enhanced performance.

The research presented here is aimed at examining whether AR information leads to objective performance improvements in terms of a lower mental load level compared with HUDs. AR information is likely to demand the driver less regarding visual, spatial resources compared with information in a HUD. Previous results have shown that AR information leads to earlier decisions in navigation tasks (cf. Bauerfeind et al., 2019). Research is needed into whether this results from a reduced mental load level with AR information. Moreover, whether a lower mental load level due to AR information eventuates in altered driving behaviour in comparison to a HUD must be analysed. This research focus is on drivers' mental load level and driving behaviour while navigating with AR information compared with a HUD. It is expected that AR information demands less mental load than HUDs, especially in ambiguous navigation situations (e.g., turning scenarios with many possible turns, which are very close to each other). In total, 59 participants had to find the target intersection in ambiguous navigation situations in an urban area. Additionally, drivers had to accomplish a non-driving-related task. This NDRT was used to measure the mental load while the driver was identifying the target intersection supported by visual navigation information presented in the AR display or in the HUD. In the study, the performance in the NDRT, driving data, and subjective data while approaching the target intersection were analysed.

2. Method

2.1. Display types for the navigation task

Fig. 1 presents the two display types (HUD and AR display) used for the navigation task. In the study, the HUD had a perceived projection

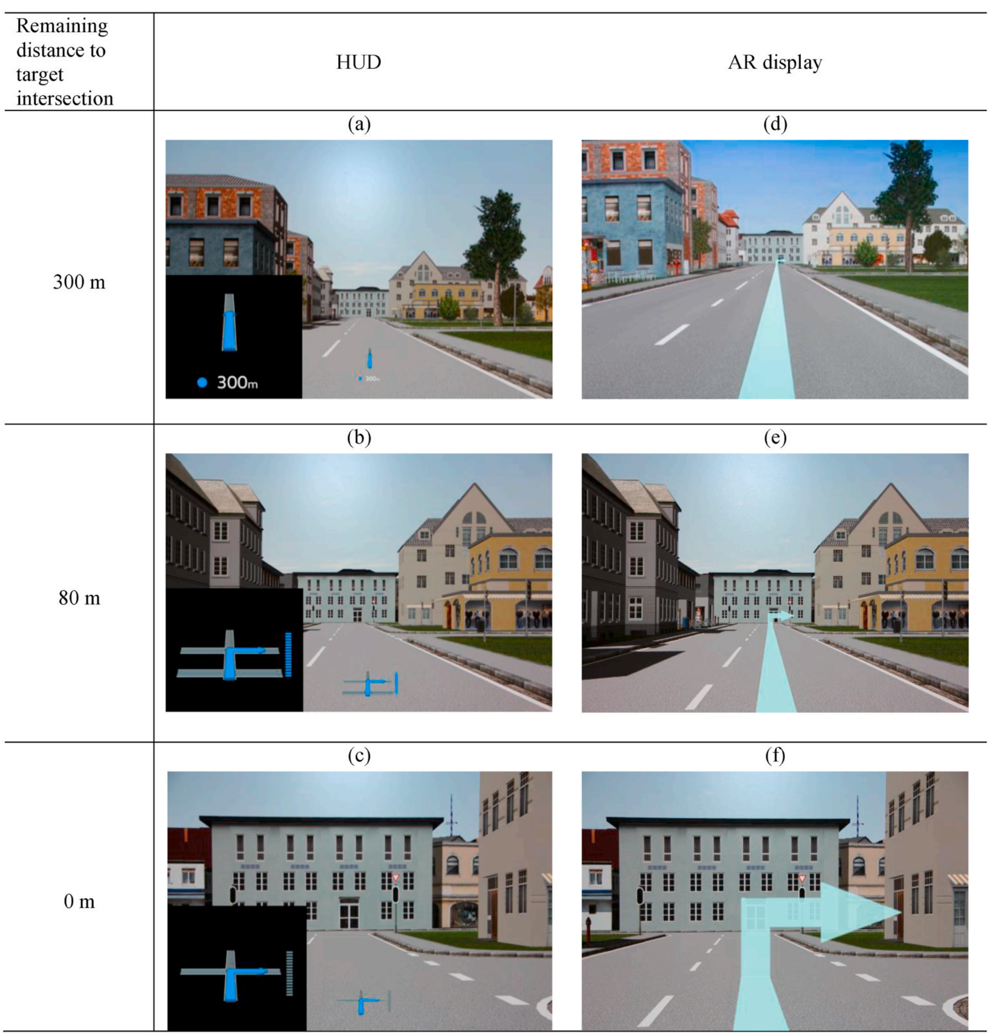

Fig. 1. Display types (HUD and AR display) presenting navigation information in respect of the remaining distance to the target intersection.

distance of approximately 1–2 m and was presented in 2D. The driver received the navigation information 300 m before the target intersection in form of a straight blue arrow (Fig. 1a). The turning direction was presented to the driver 80 m before the target intersection (Fig. 1b). The bar graph next to the blue turning arrow in the HUD indicated the remaining distance and was thus reducing while reaching the target intersection (Fig. 1b+c). With the AR display, the driver received the navigation information 300 m before the target street by way of a turquoise 3D arrow (Fig. 1d). This arrow seemed to be placed onto the driving environment. The arrow was sticking out in a 90° angle, pointing into the target intersection (Fig. 1f).

2.2. Static driving simulator

The static driving simulator with a mock-up from the Group Innovation of Volkswagen Aktiengesellschaft was used to conduct the study. This mock-up represented a vehicle with an automatic gear shift. One projection screen placed 3.8 m in front of the mock-up with a width of 3.8 m was used. This resulted in a visual field of 56° for the driver. The projector resolution was 1920 x 1200 pixels. The software Virtual Test Drive (Hexagon AB, 2021) was used to implement the simulation world. For the two display types (HUD and AR display) the software Unity was used. Besides depth cues like motion parallax, a stereo simulation provided additional realistic depth cues for the participant while observing the simulated world and the navigation information. Stereoscopy means presenting different pictures from slightly different angles of vision for the left and right eye. Hence, an impression of depth results for the participant. Therefore, participants had to wear shutter glasses.

2.3. Driving simulation, driving task, and cognitive, spatial non-driving-related task

The driving scenario used in the study involved straight roads in an urban area with 13 intersections per drive without any traffic lights. To design ambiguous navigation situations, the distance between the intersections was 40 m. According to the navigation information presented in the display (HUD or AR display), the participants' task was to find the target intersection and stop in front of it. When the target intersection was identified, participants were asked to switch off the navigation information with a button on the steering wheel while driving (cf. Bauerfeind et al., 2019). The drive was completed when the participant stopped in front of the target intersection. The experimenter gave feedback after each drive as to whether the decision was correct or incorrect. When a wrong intersection was chosen, the participants were asked to repeat the drive at the end of the HUD or the AR session. There was no other traffic in the streets. According to German regulations, the speed limit was 50 km/h.

To capture the resource demands of the ambiguous navigation task, the participants performed an NDRT in addition to the navigation task. Like the navigation task, the NDRT in this study was a cognitive and spatial task. It was used to measure the mental load while the driver was identifying the target intersection with the navigation information. For the NDRT, the participants heard numbers from one to nine via two speakers, one placed in front and one behind the simulator mock-up. The task was to evaluate every number by answering "correct" when hearing even numbers from the front speaker, as well as odd numbers from the back speaker. Participants had to say "wrong" when hearing odd numbers from the front speaker as well as even numbers from the back speaker. The interval of the numbers was 2 s.

A NDRT performance indicates changes in the primary task resource demand (Jahn et al., 2005). The NDRT performance is supposed to be inversely proportional to the performance in the primary task (Jahn et al., 2005): the more demanding the primary task is, the worse the NDRT performance gets. In contrast, the driver has more resources free to perform better in the NDRT when the primary task is less demanding. To measure drivers' mental load continuously, the NDRT in this study was a continuous highly demanding task. The premise of this method is the fact that the NDRT fits to the primary task (Jahn et al., 2005). The idea is that the primary task and the NDRT overlap to the extent that increasing the difficulty of the primary task degrades performance on the NDRT. But the NDRT should not be that demanding that it completely interferes with the primary task. Here, the primary task was navigating with the display types, which is a spatial task. The NDRT is also a spatial task as the speakers were positioned in front and behind the mock-up. Therefore, the two tasks demand the driver on the same processing channel (Wickens, 2002). However, it should be noted that the navigation task is also a visual attention task, whereas the NDRT is auditory with speech response. The speech response makes this task more realistic because it is similar to talking to car passengers.

This cognitive task was also aimed at raising the drivers' mental load, because driving on a straight urban road without other traffic was very easy. This task was to be performed verbally, not manually, because the participants had to steer and press a button when identifying the target intersection (cf. Bauerfeind et al., 2019).

2.4. Experimental design

For the driving simulator study, the APA ethical standards were strictly followed. A within-subjects design was used with participants randomly assigned to start either with the AR display or the HUD (Fig. 2). The seven drives per display type differed regarding the correct destination in the ambiguous navigation situations and were ordered randomly. Each drive was about 1 min and 30 s. The independent variable was the display type (AR display or HUD).

Fig. 3 presents the dependent variables. One dependent variable was the driver's response time in the NDRT. The endpoint for the analysis was the participant's individual response point (cf. Bauerfeind et al., 2019), which is the point when the participant identified the target intersection while driving with the AR display or the HUD by switching off the display. This was decided because it is assumed that the task demand will decrease after the response point again. Previous results showed that, when driving with the AR display, participants identified the target intersection on average 69.5 m (SD = 21.4 m) before (Bauerfeind et al., 2019). With the HUD, they needed to be closer to the target intersection (M = 39.9 m, SD = 13.2 m), which was a significant difference. Results concerning the response points are presented in Bauerfeind et al. (2019). On the basis of the response points the starting point for the analysis for the NDRT was set to 120 m before the target intersection to have a stretch for the analysis (Fig. 3).

The second and third dependent variables were driving performance. The distance to target intersection at accelerator pedal release was analysed. The accelerator pedal release is seen as the first action for the upcoming stop.

Furthermore, the approach velocity to target intersection was

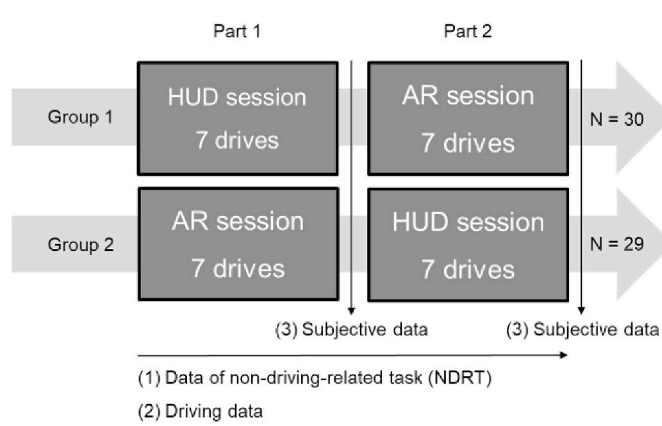

Fig. 2. Experimental setup of the driving simulator study.

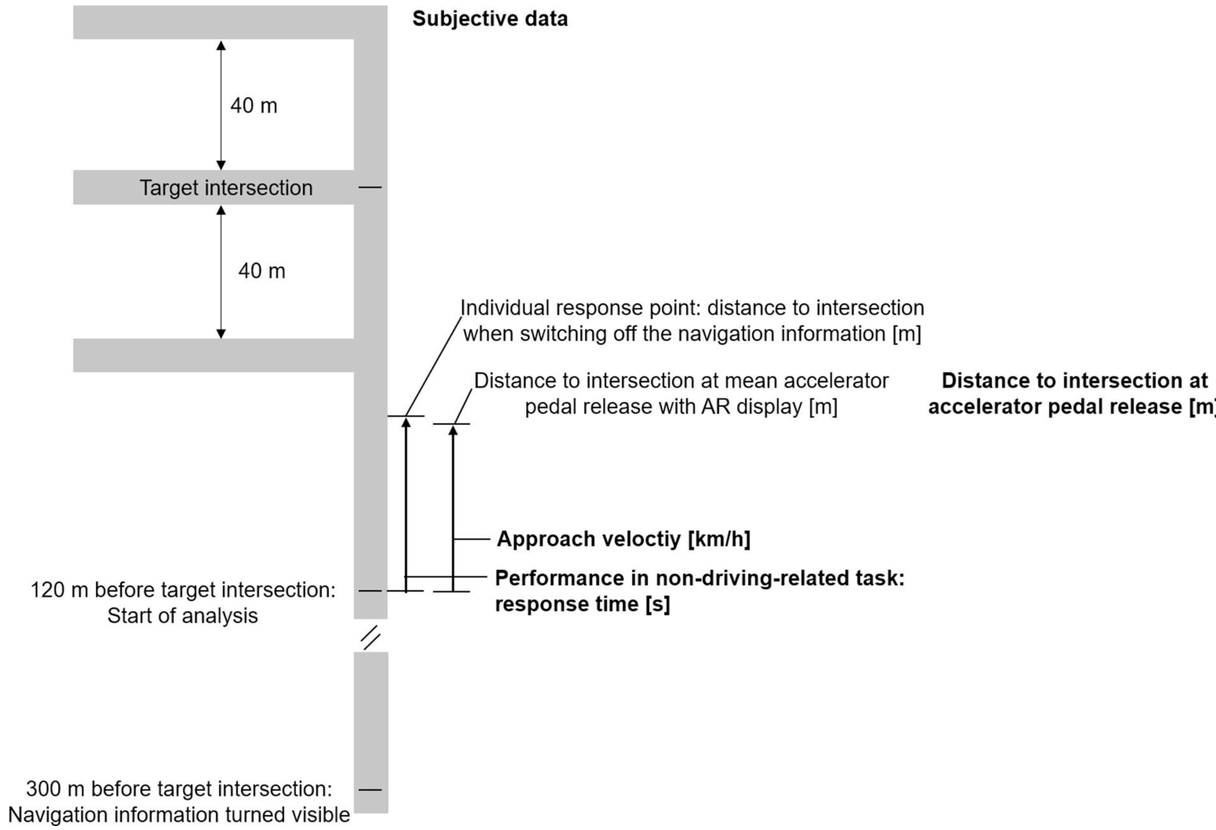

Fig. 3. Example of a drive to the target intersection with the dependent variables. Dependent variables are indicated by bold face.

examined. The starting point for the analysis of the approach velocity was 120 m before the target intersection and the endpoint was the mean accelerator pedal release for AR averaged across all participants (second dependent variable, Fig. 3). With the HUD the pedal release happened later and therefore the mean with AR was chosen as a consistent endpoint. This pedal release led to a reduced velocity and therefore the velocity after this point did not count as approach velocity anymore.

Finally, subjective ratings by the drivers were also analysed. Participants rated the display types regarding reducing mental load in the navigation task ('To what extent does the display reduce mental load in the navigation task?'). There were also subjective ratings concerning the difficulty and the effort of the driving task and the NDRT. Regarding the driving task, there were two items (difficulty of the driving task: 'How difficult was the driving task in general?', effort for the driving task: 'How demanding was the driving task in general?'). The driving task included the navigation task as well as the NDRT. Furthermore, participants were asked two items regarding the NDRT (difficulty of the NDRT: 'How difficult was the NDRT while driving?' and effort for the NDRT: 'How demanding was the NDRT while driving?'). All closed questions were answered with a 15-point rating scale (Heller, 1982) (Fig. 4).

2.5. Participants

A total of 61 drivers participated in the study. Because of simulator sickness and lack of 3D-vision data of 59 participants could be analysed (n = 18 female). The participants averaged 37.1 years old (*SD* = 11.1 years) and were recruited for the driving simulator study from the test driver pool of Volkswagen Group Innovation. On average they drove 18274.7 km per year (*SD* = 11368.9 km). 44% of the participants stated they had experience with a HUD, but the majority of those stated that they rarely use it. 58% of the participants used vision correction by glasses or contact lenses. To get to know the display types and the simulator, participants started with a training. All participants received a present for their participation.

2.6. Procedure

After being welcomed, the participants completed the socio-demographic questions. Then, they were tested for stereo vision by applying the Butterfly Stereo Acuity Test (VAC Vision Assessment Corporation, 2007) and for colour vision performance. The experimenter explained the primary task (the navigation task) as well as the NDRT (the auditory cognitive spatial task) to the participants. They received instructions regarding the buttons on the steering wheel for switching off the display. To experience the simulation in 3D, participants wore shutter glasses. They started with training the navigation task and NDRT without driving. Then, they practiced both tasks simultaneously until they could perform without errors. After the training session, the experiment started with seven drives, either with the AR display or the HUD. Participants were randomly assigned to one of the orders. If the participants identified the wrong intersection, they had to repeat the drive at the end of the respective display session. After finishing the display sessions, they rated the two display types using the evaluation

very easy			easy			moderate			difficult			very difficult		
-	0	+	-	0	+	-	0	+	-	0	+	-	0	+

Fig. 4. 15-point rating scale according to Heller (1982) with regard to the question: How difficult was the NDRT while driving? This scale was used for all closed questions.

questionnaire. After the experiment the participants received a present. The experiment lasted a total of 130 min.

3. Results

3.1. Data preparation

In total, 59 participants were analysed. The NDRT was used to measure the drivers' mental load. Because of technical issues, 697 trials could be observed in this analysis. To prepare the NDRT data, speech recognising software was written in R to extract the response time. For the analysis, the response time is the time stamp when participants started answering. A hierarchical linear model with maximal random effects structure and models with reduced random effects structures (Barr et al., 2013) were fitted and AIC-values were compared. The model, which scored the lowest on AIC-values explained the data best and was therefore chosen for the analysis. Resulting a hierarchical linear model with varying intercept and varying slope was analysed (N = 59). The dependent variable was mean response time in seconds. The independent variables were display type (AR display, HUD) and remaining distance to the target intersection in meters. Participants were analysed as a random intercept.

With regard to the driving data, a *t*-test for dependent samples was used to analyse the distance to the target intersection at accelerator pedal release (N = 59). To analyse the approach velocity to the target intersection, a hierarchical linear model with varying intercept and varying slope was modelled (N = 59). The dependent variable was mean velocity to the target intersection in kilometers per hour. Participants were analysed as a random intercept.

Subjective data of 59 participants were analysed using repeated measures Analysis of Variance (rmANOVA) to examine the effects of display type (HUD vs. AR display) and the order of the display type as between subject factor.

3.2. Non-driving-related task (NDRT)

Fig. 5 shows line graphs of all predicted response times for the NDRT of the participants while navigating to the target intersection with the AR Display (Fig. 5, left) and the HUD (Fig. 5, right). Each line presents all drives per display type for the respective participant. The black lines present the predicted mean response times for the respective display type. The NDRT response time increased with distance to target for the HUD, but not for the AR display. Fig. 5 and Table 1 show the closer participants got to the target intersection with the HUD, the slower they responded to the NDRT ($t = 4.19$, $p < .001$).

The hierarchical linear model predicts a significant difference in response times between the AR display and the HUD when reaching the target intersection (distance to target intersection = 0 m, Table 1). With the AR display, participants will answer on average after 1.35 s the moment they reach the target intersection, whereas with the HUD the value is 1.66 s ($t = 4.11$, $p < .001$). This is 0.31 s longer (Table 1).

3.3. Driving data

Regarding the driving data, there was a significant difference between the two display types in regard of the distance to the target intersection at accelerator pedal release (Fig. 6). With the AR display, participants released the accelerator pedal 70.2 m ($SD = 15.6$ m) on average before the intersection, whereas with the HUD the value was 59.3 m ($SD = 10.3$ m), ($t(58) = 6.37$, $p < .001$, $r = .64$) (Fig. 6).

Fig. 7 shows the approach velocity to the target intersection with the AR display and the HUD. The participants had a steady approach velocity with both display types. With regard to the difference in approach velocity between the display types, participants drove significantly slower with the HUD (Fig. 7, Table 2). At 120 m before the target intersection participants drove 52.26 km/h with the AR display, whereas with the HUD the velocity was 49.35 km/h ($t = -70.23$, $p < .001$; Table 2). Thus, 2.91 km/h slower (Table 2).

Furthermore, the hierarchical linear model shows a significant difference in acceleration behaviour ($t = 8.14$, $p < .001$). However, this difference is so small that it does not show a practical relevance.

3.4. Subjective evaluation

The subjective evaluation of reduced mental load in the ambiguous navigation task showed a significant main effect between the display types ($F(1, 57) = 130.39$, $p < .001$, $\eta_p^2 = .70$). Participants rated that the AR display reduced more mental load in the navigation task than the HUD (AR display: $M = 11.2$, $SD = 2.2$; HUD: $M = 7.3$, $SD = 2.4$) (Fig. 8). The interaction between display type and order of display type was not significant ($F(1, 57) = 1.33$, $p > .10$, $\eta_p^2 = .02$) (cf. Bauerfeind et al., 2019).

According to Fig. 9 (a+b), participants rated the difficulty and the effort regarding the driving task significantly higher when driving with the HUD than with the AR display. Here, the driving task included both the navigation task and the NDRT. There was a significant main effect between the displays in regard to the difficulty of the driving task ($F(1, 57) = 47.42$, $p < .001$, $\eta_p^2 = .45$). Thus, participants rated the driving task as less difficult when driving with the AR display than with the HUD (AR display: $M = 7.3$, $SD = 2.3$; HUD: $M = 9.3$, $SD = 2.2$) (Fig. 9a). The interaction between display type and order of display type was not significant ($F(1, 57) = 2.05$, $p > .10$, $\eta_p^2 = .04$).

There was also a significant main effect concerning the effort of the driving task ($F(1, 57) = 70.27$, $p < .001$, $\eta_p^2 = .55$). While driving with the AR display, participants evaluated the driving task as less

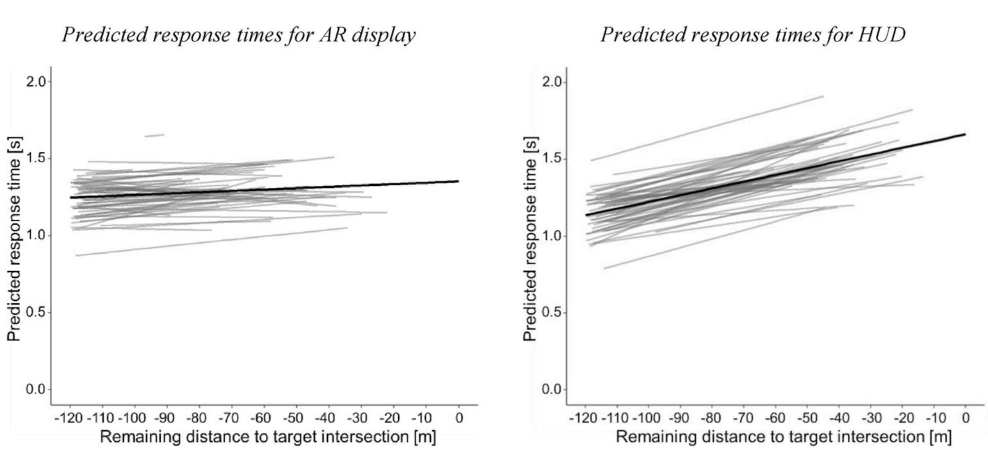

Fig. 5. Line graphs of all predicted response times in the NDRT in seconds (N = 59) driving with the AR display (left) and the HUD (right) to the target intersection (0 m). Each line presents all drives per display type for the respective participant. Starting point for the analysis was 120 m before the target intersection. Ending point for the analysis was the individual response point when the participant identified the target intersection, which is why lines differ in length. The black lines show the respective predicted mean response times.

Table 1
Hierarchical linear model analysis with predicted mean response times as the dependent variable and participant as a random intercept. The predicted mean response times refer to the moment reaching the target intersection.

Measures	Estimate	*SE*	*t*	*p*	95% CI			Estimate	*SD*
(constant)	[s]				low	high			
Fixed effects							**Random effects**		
(Intercept)*	1.35	0.07	18.59	<.001	1.21	1.50	(Intercept)	0.07	0.26
HUD	**0.31**	**0.08**	**4.11**	**< .001**	**0.16**	**0.46**			
Distance to target intersection [m]	0.0009	0.0008	1.14	>.10	−0.0006	0.002	Distance to target intersection	0.000006	0.003
HUD * Distance to target inter-section [m]	**0.004**	**0.0008**	**4.19**	**< .001**	**0.002**	**0.005**	Residuals	0.13	0.36

Note. *predicted response time with the AR display at the target intersection (0 m). *SE* = standard error. *SD* = standard deviation. R^2marginal = 0.05, R^2conditional = 0.19. Significant effects are indicated by bold face, N = 59.

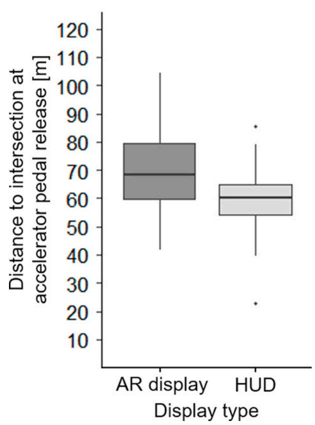

Fig. 6. Boxplots of the distance to target intersection at accelerator pedal release of all participants regarding the two display types (N = 59).

demanding in comparison to the HUD (AR display: $M = 7.2$, $SD = 2.1$; HUD: $M = 9.4$, $SD = 2.1$) (Fig. 9b). A hybrid interaction effect between display type and order of display type points out that the main effect concerning the display type is globally interpretable ($F(1, 57) = 9.08$, $p < .05$, $\eta_p^2 = .14$).

Regarding the NDRT, participants rated the difficulty and the effort of the NDRT significantly higher when driving with the HUD (Fig. 9c+d). Concerning the difficulty of the NDRT, there was a significant main effect ($F(1, 57) = 14.52$, $p < .001$, $\eta_p^2 = .20$). Participants rated the NDRT with both display types as "moderate difficult" but less difficult with the AR display (AR display: $M = 7.6$, $SD = 2.5$; HUD: $M = 8.8$, $SD = 2.7$) (Fig. 9c). A hybrid interaction effect between display type and order of display type points out that the main effect concerning the display type is globally interpretable ($F(1, 57) = 4.39$, $p < .05$, $\eta_p^2 = .07$).

There was also a significant main effect concerning the effort of the NDRT between the two display types ($F(1, 57) = 29.75$, $p < .001$, $\eta_p^2 = .34$). Driving with the AR display, participants rated the NDRT less demanding than with the HUD (AR display: $M = 7.8$, $SD = 2.6$; HUD: $M = 9.3$, $SD = 2.6$) (Fig. 9d). A hybrid interaction effect between display type and order of display type points out that the main effect concerning the display type is globally interpretable ($F(1, 57) = 10.49$, $p < .05$, $\eta_p^2 = .16$).

4. Discussion

The aim of the driving simulator study was to examine differences in drivers' mental load in a navigating task using an AR display or a HUD. The study presented here confirms that AR information is more easily and quickly understood by the driver than a HUD. Results showed that AR information led to more confident driving behaviour than with the HUD. Furthermore, data revealed that AR information requires less of the drivers' mental load in understanding and interpreting compared with HUDs. This was shown with objective performance measures in a NDRT and subjective ratings. It seems that the proximity of virtual and

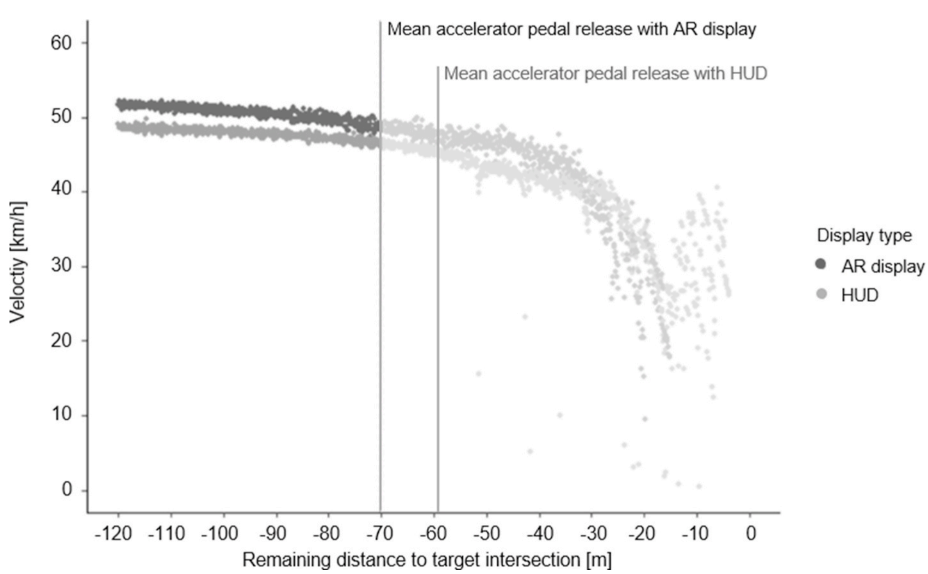

Fig. 7. Graph of approach velocity of all participants (N = 59) with the two display types, HUD and AR display, starting from 120 m before the target intersection to the moment of mean accelerator pedal release with AR (70.2m before the target intersection, marked by the left vertical line. Mean accelerator pedal release with HUD was at 59.3m before the target intersection, marked by the right vertical line).

Table 2
Hierarchical linear model analysis with mean velocity as the dependent variable from 120 m to 70.2 m before the target intersection.

Measures (constant)	Estimate [km/h]	*SE*	*t*	*p*	95% CI low	95% CI high
Fixed effects						
(Intercept)*	52.26	0.54	94.41	<.001	51.18	53.35
HUD	**−2.91**	**0.04**	**−70.23**	**< .001**	**−2.99**	**−2.83**
Distance to target intersection [m]	**−0.05**	**0.00**	**−47.37**	**< .001**	**−0.05**	**−0.05**
HUD * Distance to target intersection [m]	**0.01**	**0.00**	**8.14**	**< .001**	**0.01**	**0.01**

Note. *velocity with AR display at −120 m before the target intersection. *SE* = standard error. *SD* random effects = 4.25 m. R^2marginal = 0.06, R^2conditional = 0.55. Significant effects are indicated by bold face, N = 59.

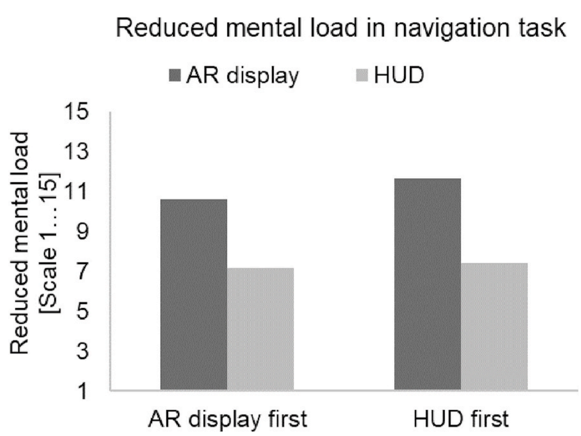

Fig. 8. Bar graphs of subjective evaluation regarding reduced mental load in the navigation task (N = 59), (1: low reduction, 15: high reduction).

real objects results in the driver not having to switch between the relevant navigation information and the real traffic situation, which is in line with assertions in the literature (Kim and Dey, 2009). As a result, AR information leads to reduced mental load.

Results of the study showed that participants responded to the NDRT faster while driving with the AR display compared with the HUD. Furthermore, participants had an almost constant response time with the AR display while approaching the ambiguous target intersection. The closer the participants with the HUD drove to the target intersection, the response times increased constantly. It seems that the closer they got, the more they concentrated on the HUD to solve the ambiguous navigation task, which therefore interfered with the NDRT. This performance loss in the NDRT indicates a raised driver's mental load through the HUD compared with the AR display. Overall, navigating with the AR display is less demanding than with the HUD and fewer resources appear necessary. Any difference between the two display types could have produced this result, but it is assumed that it was caused by the superimposition of AR information on the respective real object, resulting in an intuitive and fast understanding being possible for the driver. Especially in navigation, following AR information demands less mental load than a HUD. Because of the superimposition of AR information, the driver does not have to map the AR information onto the environment. Resulting the assertion in the literature can be confirmed (Pfannmüller, 2017).

The subjective data confirmed the reduced mental load while driving with AR. The participants rated that the AR display reduced more mental load in the navigation task than the HUD. They stated that the NDRT was more difficult and more demanding when driving with the HUD. Therefore, in line with previous assertions (Bengler et al., 2015; Israel, 2012; Kim and Dey, 2009; Pfannmüller, 2017; Pfannmüller et al., 2015), mental effort appears to be reduced with AR information relative to the HUD. Furthermore, the drivers rated the driving task including the navigation task and the NDRT as less demanding and less difficult with AR than with the HUD. This confirms that, compared with the HUD, navigating, and passing the NDRT is easier with AR.

Regarding the driving data, results showed that participants took their foot off the accelerator pedal significantly earlier with AR than with a HUD. Taking the foot off is the first action for the upcoming stop. With the AR display, participants identified the target intersection earlier (cf. Bauerfeind et al., 2019) and thus showed a first action for the upcoming stop earlier than with the HUD. Also, participants had to stop acceleration earlier to be able to stop at the target intersection because the analysis of the approach velocity showed they drove faster with AR than with the HUD. The difference in velocity between the two display types was only 3 km/h, but very consistent. Participants intuitively followed the AR arrow lying on the street, which might have led to more confident driving behaviour than with the HUD. With the HUD, they had to map the information to the real world to identify the correct target intersection. A reduced velocity might therefore show compensating behaviour. This compensating behaviour is described in the speed-accuracy tradeoff (cf. Garrett, 1922; Heitz, 2014; Woodworth, 1899). Accordingly, it takes longer to accomplish a task more accurately, but saving time results in reduced accuracy. Driving slower might have allowed more time to choose the target intersection with the HUD.

Regarding the limitations of the study, it has to be noted that the information in the two displays were not informationally equivalent at any time. At the beginning of each drive, the HUD presented the information to go straight. The turning direction could be presented when there were fewer than three intersections to come. In contrast, the driver could see from far away whether it was a left or a right turn because of the superimposition of AR information on the real street. Therefore, this might be an advantage of AR information.

Concerning the analysis of the NDRT, it has to be considered that it is based on the range from 120 m before the target intersection to the individual response point. Thus, the intercept, which refers to the moment the participants reached the target intersection, is an estimate. However, with this hierarchical linear model, the rise in response times approaching the target intersection could be modelled, which makes the model very suitable for the analysis.

Furthermore, the endpoint for the analysis of approach velocity was the mean accelerator pedal release with the AR display. Lifting the foot off the accelerator pedal was seen as the first action for the upcoming stop. Therefore, the velocity after this point did not count as approach velocity anymore. However, a few participants lifted their foot off the accelerator pedal earlier than this mean, therefore already reducing speed. Yet the velocity with the AR display was still higher than with the HUD. This means that the counterargument even strengthens the difference in velocity between the two display types.

Overall, deviations with respect to handling the driving simulator have to be considered. The velocity is often underestimated in a simulator. Deviations between the expected and the driven velocity can result in enhanced braking and acceleration.

In the study presented here, in order to achieve a high standardization for the ambiguous navigation situations, there was no other traffic in the streets. Thus, a systematic evaluation of subjective and objective data could be made. Because the AR information seemed to be placed onto the driving situation, the interrelatedness between this information and other traffic has to be tested. For further research on AR, it is necessary to evaluate these findings in a real driving environment. In particular, effects of traffic density on the interaction with AR information have to be examined. Putting information in the forward field of view could occlude information that might be important for driving (e.g., a pedestrian, a bicyclist).

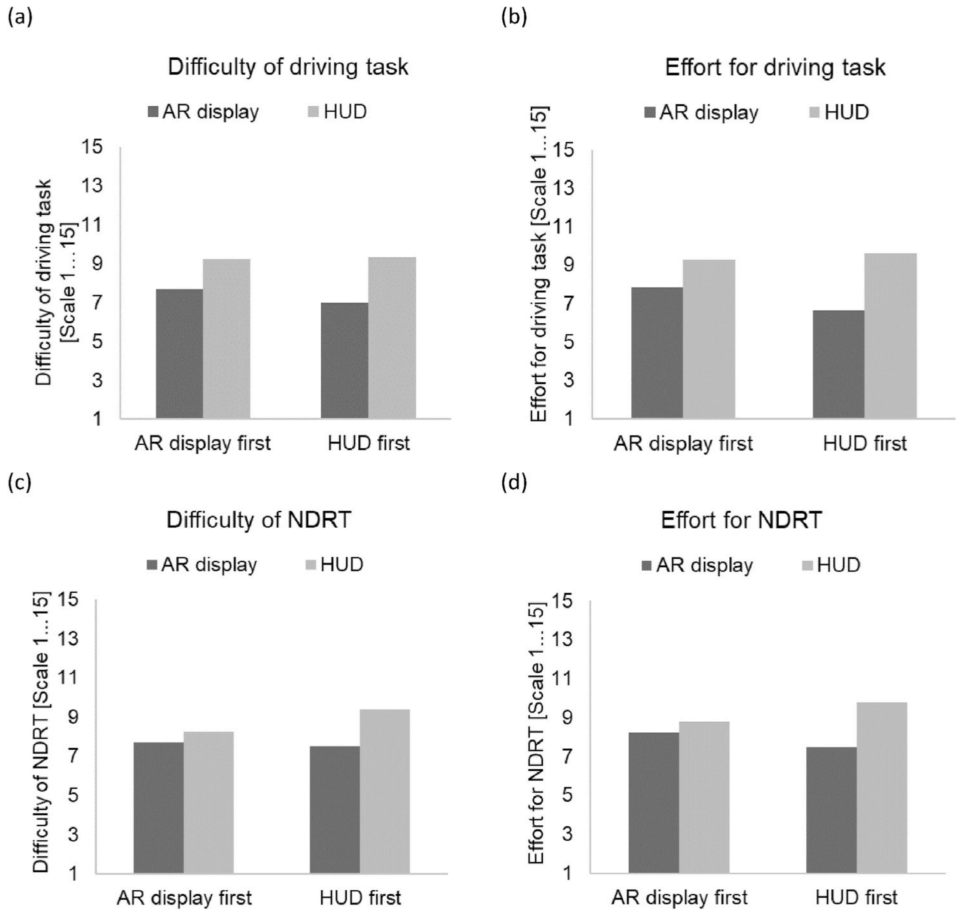

Fig. 9. Bar graphs of subjective evaluation regarding difficulty and effort of the driving task (navigation task & NDRT) and the NDRT (N = 59), (1: low consent, 15: high consent on the specific item).

5. Conclusion

This research was focused on the effects of AR information on drivers' mental load in a navigation task. Overall, this study confirms that AR information is more easily and quickly understood by the driver than a HUD. Moreover, data showed that AR information led to more confident driving behaviour than with the HUD. Furthermore, fewer resources appear necessary while navigating with AR information. The reduced mental load level might result in safer driving with less inattention because the driver is able to spend the necessary resources on the driving task itself. In this study, stereo simulation was used so participants could experience the simulated world and the navigation information in 3D. Thus, participants got a more realistic impression of AR depth cues for navigation. In this study, AR information was perfectly implemented in the driving scenarios. Further research should evaluate whether deviations in positioning might incur costs resulting in a higher mental load level in return.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We express our sincere thanks to Michael Wittkämper for implementing the AR display in the simulated world, to Lea Dingler for her support to collect the data and to Anne-Marie Wesseling for her support in data preparation. Gratefully acknowledged is Professor Stephen Arnott, who provided feedback on the manuscript.

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